SearchOS: Agents Share Evidence, Gaps, and Failed Searches

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SearchOS is an MIT-licensed system for research questions that require filling many related facts into tables. Instead of leaving progress inside each agent's chat history, its shared search state records unfinished cells, source-backed evidence, coverage gaps, conflicts, and failed approaches. Up to eight agents search in parallel, while a continuous scheduler sends new work to any free slot rather than waiting for the whole group. In the authors' evaluation using

GLM-5 for agent roles and Qwen3.5-35B-A3B for evidence extraction, SearchOS reached 80.3 item F1 on WideSearch, versus 76.0 for the strongest listed baseline. It scored 76.5 set F1 on GISA, versus 63.1. Both headline results are Max@3, the best of three attempts per case, not typical single-run performance.

That shared memory helps with jobs such as comparing universities, products, or regulations across dozens of fields. One agent can leave a cited fact or a blocked route, and the next can target the missing cell instead of repeating the same dead search. On the same 100 WideSearch questions, the paper's combined skill system cut session time 36.6%, search calls 39.1%, and page calls 42.7%, while item F1 rose from 78.3 to 80.3. The public alpha runs in a terminal or browser and supports several model and search providers, but it requires Python 3.11, provider credentials, and paid API usage in many configurations. The authors do not report cost per completed table or single-run score distributions, so efficiency and reliability still need independent testing.

Read More: OpenSeeker trains a search agent from 11,700 public examples.

Sources:

- SearchOS-V1 paper (arXiv)
- SearchOS source code and setup guide (GitHub)
- WideSearch benchmark paper (arXiv)
- GISA benchmark paper (arXiv)

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